

Ke Xue

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Education

Nanjing University

Sep. 2019 – Sep. 2026 (expected)

Ph.D. candidate in Computer Science, School of Artificial Intelligence, Advisor: Chao Qian. Nanjing, China.

Sun Yat-Sen University

Sep. 2015 – Jun. 2019

B.Sc. in Mathematics and Applied Mathematics, School of Mathematics. Guangzhou, China

Research Interests

Black-box optimization algorithms—including evolutionary algorithms, Bayesian optimization, and learning to optimize—and their applications.

Publications and preprints

* indicates equal contribution; † corresponding author.

Journal articles

- Zhengbo Lu*, Ke Xue*, Yiying Deng*, Junxuan Fan, Peiyue Fang, Bridget S. Wade, Laia Alegret, Michael J. Benton, Yuchang Wu, Chao Qian, Xudong Hou, Yukun Shi, Peter M. Sadler, Huiqing Xu, Zhi-Hua Zhou, and Shuzhong Shen. Complex Marine Ecological Response during the Eocene-Oligocene Revealed by Global Foraminiferal Record. *Nature Communications*, 2026.
- Zhengbo Lu, Junxuan Fan, Bridget S. Wade, Paul N. Pearson, Yukun Shi, Chao Qian, Ke Xue, Jiao Yang, Yujie Shi, and Muchen Lai. Graph-Optimized Graphic Correlation: An Automated Framework for Combining Stratigraphical Data. *Lethaia*, 2026, 60(1): 1–16.
- Han Hu*, Ke Xue*, Yishen Sun, Qing Zhu, Hans K. Carlson, Ruiwen Hu, Rong-Xi Tan, Chao Qian, Weigen Huang, Jizhong Zhou, Jingdong Mao, Thomas W. Crowther, Zhi-Hua Zhou, Jiabao Zhang, and Yuting Liang. Reducing the Discrepancy in Quantifying the Temperature Dependence of Global Wetland Methane Emission. *Global Change Biology*, 2026, 32(2): e70748.
- Tianyi Chu, Hanhui Huang, Junxuan Fan, Yiying Deng, Tao Xu, Chao Qian, Ke Xue, H. David Sheets, Michael H. Stephenson, Yukun Shi, and Xudong Hou. HORSE: Harmonize Regional and Global Stratigraphic Records through Horizon Sequencing. *Palaeogeography, Palaeoclimatology, Palaeoecology*, 2025, 670: 112976.
- Ren-Jian Wang*, Ke Xue*, Yutong Wang, Peng Yang, Haobo Fu, Qiang Fu, and Chao Qian. Diversity from Human Feedback. *Frontiers of Computer Science* (see also NeurIPS 2023 workshop version).
- Zhengbo Lu, Junxuan Fan, Bridget S. Wade, James Ogg, Laia Alegret, Peter M. Sadler, Michael H. Stephenson, Yukun Shi, Chao Qian, Ke Xue, and Peiyue Fang. Improving the Temporal Resolution of Middle Eocene–Late Oligocene Foraminiferal Biomagneto-Chronology: Insights from CONOP and Chronologic Significance of Biotic Events. *Palaeogeography, Palaeoclimatology, Palaeoecology*, 2025, 669: 112929.
- Han Hu*, Chao Qian*, Ke Xue*, Rainer Georg Jörgensen, Marco Keiluweit, Chao Liang, Xuefeng Zhu, Ji Chen, Yishen Sun, Haowei Ni, Jixian Ding, Weigen Huang, Jingdong Mao, Rong-Xi Tan, Jizhong Zhou, Thomas W. Crowther, Zhi-Hua Zhou, Jiabao Zhang, and Yuting Liang. Reducing the Uncertainty in Estimating Soil Microbial Derived Carbon Storage. *Proceedings of the National Academy of Sciences*, 2024, 121(35): e2401916121.
- Ke Xue, Yutong Wang, Cong Guan, Lei Yuan, Haobo Fu, Qiang Fu, Chao Qian, and Yang Yu. Heterogeneous

Multi-Agent Zero-Shot Coordination by Coevolution. *IEEE Transactions on Evolutionary Computation*.

- Cong Guan, Feng Chen, Ke Xue, Chunpeng Fan, Lichao Zhang, Ziqian Zhang, Pengyao Zhao, Zongzhang Zhang, Chao Qian, Lei Yuan, Yang Yu. One by One, Continual Coordinating with Humans via Hyper-Teammate Identification. *Transactions on Machine Learning Research*.
- Cong Guan, Ke Xue, Chunpeng Fan, Feng Chen, Lei Yuan, Chao Qian, and Yang Yu. Open and Real-World Human-AI Coordination by Heterogeneous Training with Communication. *Frontiers of Computer Science*, 2025, 19(4): 194314.

Conference papers

- Ming Chen, Sheng Tang, Rong-Xi Tan, Ziniu Li, Jiacheng Chen, Ke Xue[†], and Chao Qian. Beyond Token-level Supervision: Unlocking the Potential of Decoding-based Regression via Reinforcement Learning. In *ICML*, 2026.
- Ruo-Tong Chen, Ke Xue, Chengrui Gao, Yunqi Shi, Tian Xu, Peng Xie, Siyuan Xu, Mingxuan Yuan, Chao Qian, and Zhi-Hua Zhou. How Can Reinforcement Learning Achieve Expert-level Placement? In *DAC*, 2026.
- Peng Xie, Ke Xue, Yunqi Shi, Ruo-Tong Chen, Chengrui Gao, Siyuan Xu, Chenjian Ding, Mingxuan Yuan, and Chao Qian. FlowPlace: Flow Matching for Chip Placement. In *DAC*, 2026.
- Chengrui Gao, Yunqi Shi, Ke Xue[†], Ruo-Tong Chen, Siyuan Xu, Mingxuan Yuan, Chao Qian, and Zhi-Hua Zhou. Expertise Can Be Helpful for Reinforcement Learning-based Macro Placement. In *ICLR*, 2026.
- Chen Lu, Ke Xue[†], Ruo-Tong Chen, Yunqi Shi, Siyuan Xu, Mingxuan Yuan, Chao Qian, and Zhi-Hua Zhou. Dynamic Algorithm Configuration for Global Placement. In *DATE*, 2026.
- Ruo-Tong Chen, Chengrui Gao, Siyuan Xu, Ke Xue, Yunqi Shi, Xi Lin, Mingxuan Yuan, Chao Qian, and Zhi-Hua Zhou. Timing-driven Detailed Placement via TimingMask-guided Path-level Optimization. In *DATE*, 2026.
- Wanqi Ren, Chengrui Gao, Yunqi Shi, Mingzhou Fan, Siyuan Xu, Ke Xue, Chenjian Ding, Mingxuan Yuan, and Chao Qian. Reinforcement Learning for Hybrid Bonding Terminal Legalization in 3D ICs. In *DATE*, 2026 (Extended Abstract).
- Chen Lu, Ke Xue[†], Lei Yuan, Yao Wang, Yaoyuan Wang, Sheng Fu, and Chao Qian. Sequential Multi-Agent Dynamic Algorithm Configuration. In *NeurIPS*, 2025.
- Yawen Zhou, Ke Xue, and Chao Qian. MGDA/D: A Multiple Gradient Descent Algorithm Based on Decomposition. In *GECCO*, 2025 (poster).
- Rong-Xi Tan, Ming Chen, Ke Xue[†], Yao Wang, Yaoyuan Wang, Sheng Fu, and Chao Qian. Towards Universal Offline Black-Box Optimization via Learning Large Language Model Embeddings. In *ICML*, 2025.
- Chengrui Gao, Haopu Shang, Ke Xue, and Chao Qian. Neural Solver Selection for Combinatorial Optimization. In *ICML*, 2025.
- Lei Song, Chenxiao Gao, Ke Xue, Chenyang Wu, Dong Li, Jianye Hao, Zongzhang Zhang, and Chao Qian. Reinforced In-Context Black-Box Optimization. In *IJCAI*, 2025.
- Yunqi Shi, Xi Lin, Siyuan Xu, Shixiong Kai, Ke Xue, Mingxuan Yuan, Chao Qian, and Zhi-Hua Zhou. ReMaP: Macro Placement by Recursively Prototyping and Periphery-Guided Relocating. In *DAC*, 2025.
- Rong-Xi Tan, Ke Xue, Shen-Huan Lyu, Haopu Shang, Yao Wang, Yaoyuan Wang, Sheng Fu, and Chao Qian. Offline Model-Based Optimization by Learning to Rank. In *ICLR*, 2025.
- Erlong Liu, Yu-Chang Wu, Xiaobin Huang, Chengrui Gao, Ren-Jian Wang, Ke Xue, and Chao Qian. Pareto Set Learning for Multi-Objective Reinforcement Learning. In *AAAI*, 2025 (Oral).
- Yunqi Shi, Siyuan Xu, Shixiong Kai, Xi Lin, Ke Xue, Mingxuan Yuan, and Chao Qian. Timing-Driven Global Placement by Efficient Critical Path Extraction. In *DATE*, 2025. **Best Paper Award**.
- Ke Xue, Ruo-Tong Chen, Xi Lin, Yunqi Shi, Shixiong Kai, Siyuan Xu, and Chao Qian. Reinforcement Learning Policy as Macro Regulator Rather than Macro Placer. In *NeurIPS*, 2024.
- Shukuan Wang*, Ke Xue*, Lei Song, Xiaobin Huang, and Chao Qian. Monte Carlo Tree Search based Space Transfer for Black Box Optimization. In *NeurIPS*, 2024. Spotlight.

- Ke Xue*, Rong-Xi Tan*, Xiaobin Huang, and Chao Qian. Offline Multi-Objective Optimization. In *ICML*, 2024.
- Ren-Jian Wang, Ke Xue, Cong Guan, and Chao Qian. Quality-Diversity with Limited Resources. In *ICML*, 2024.
- Chao Qian, Ke Xue, and Ren-Jian Wang. Quality-Diversity Algorithms Can Provably Be Helpful for Optimization. In *IJCAI*, 2024.
- Chengrui Gao, Haopu Shang, Ke Xue, Dong Li, and Chao Qian. Towards Generalizable Neural Solvers for Vehicle Routing Problems via Ensemble with Transferrable Local Policy. In *IJCAI*, 2024.
- Ke Xue*, Xi Lin*, Yunqi Shi, Shixiong Kai, Siyuan Xu, and Chao Qian. Escaping Local Optima in Global Placement. In *DAC*, 2024 (Work-in-Progress poster).
- Ke Xue*, Ren-Jian Wang*, Pengyi Li, Dong Li, Jianye Hao, and Chao Qian. Sample-Efficient Quality-Diversity by Cooperative Coevolution. In *ICLR*, 2024. Spotlight.
- Xiaobin Huang, Lei Song, Ke Xue, and Chao Qian. Stochastic Bayesian Optimization with Unknown Continuous Context Distribution via Kernel Density Estimation. In *AAAI*, 2024.
- Yunqi Shi, Ke Xue, Lei Song, and Chao Qian. Macro Placement by Wire-Mask-Guided Black-Box Optimization. In *NeurIPS*, 2023. Spotlight. 21st ACM SIGEVO Humies Bronze Award.
- Ren-Jian Wang, Ke Xue, Haopu Shang, Chao Qian, Haobo Fu, and Qiang Fu. Multi-objective Optimization-based Selection for Quality-Diversity by Non-surrounded-dominated Sorting. In *IJCAI*, 2023.
- Ziqian Zhang, Lei Yuan, Lihe Li, Ke Xue, Chengxing Jia, Cong Guan, Chao Qian, and Yang Yu. Fast teammate adaptation in the presence of sudden policy change. In *UAI*, 2023.
- Lei Yuan, Zi-Qian Zhang, Ke Xue, Hao Yin, Feng Chen, Cong Guan, Li-He Li, Chao Qian, and Yang Yu. Robust multi-agent coordination via evolutionary generation of auxiliary adversarial attackers. In *AAAI*, 2023. Oral.
- Ke Xue*, Jiacheng Xu*, Lei Yuan, Miqing Li, Chao Qian, Zongzhang Zhang, and Yang Yu. Multi-Agent Dynamic Algorithm Configuration. In *NeurIPS*, 2022. Spotlight.
- Lei Song*, Ke Xue*, Xiaobin Huang, and Chao Qian. Monte Carlo Tree Search based Variable Selection for High Dimensional Bayesian Optimization. In *NeurIPS*, 2022. Spotlight.
- Yutong Wang*, Ke Xue*, and Chao Qian. Evolutionary Diversity Optimization with Clustering-based Selection for Reinforcement Learning. In *ICLR*, 2022.
- Ke Xue, Chao Qian, Ling Xu, and Xu-Dong Fei. Evolutionary Gradient Descent for Non-convex Optimization. In *IJCAI*, 2021.
- Chao Qian, Hang Xiong, and Ke Xue. Bayesian Optimization using Pseudo-Points. In *IJCAI*, 2020.

Service

- **Conference reviewer:** *ICML*, *NeurIPS*, *ICLR*, *AAAI*, *ECAI*, *AutoML*.
- **Journal reviewer:** *SCIENCE CHINA Information Sciences*; *IEEE TPAMI*; *IEEE TEvC*; *IEEE TETCI*; *Swarm and Evolutionary Computation*; *Frontiers of Computer Science*.

Awards

- DATE 2025 Best Paper Award (with Yunqi Shi, Siyuan Xu, Shixiong Kai, Xi Lin, Mingxuan Yuan, and Chao Qian), 2025.
- Top-20 nomination for the Baidu Scholarship, 2025.
- Young Elite Scientists Sponsorship Program by CAST for Ph.D. students, 2024.
- National Science Foundation of China grant for Ph.D. students, 2024.
- National Scholarship, Ministry of Education of China, 2024.
- 21st ACM SIGEVO Humies Bronze Award (with Yunqi Shi, Lei Song, and Chao Qian), 2024.
- National Scholarship, Ministry of Education of China, 2021.